

MASTER'S THESIS IN PHYSICS

Integration of an SiPM-Based Scintillation Detector into the Phase-2 CMS Drift Tube Readout Chain Using the lpGBT ASIC

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Abstract

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Contents

1	Introduction	1
2	The Standard Model of particle physics and cosmic muons	2
2.1	The Standard Model of particle physics	2
2.1.1	Matter particles	2
2.1.2	Force mediators	2
2.1.3	Higgs boson	2
2.1.4	Incompleteness and challenges	2
2.2	Cosmic muons	3
2.2.1	Cosmic radiation	3
2.2.2	Atmospheric creation of muons	3
3	Principles of particle detection	4
3.1	Interaction of particles and matter	4
3.1.1	Photon interactions	4
3.1.2	Charged particle interactions	4
3.2	Scintillators and photon detection	4
3.2.1	Scintillators	4
3.2.2	Silicon photomultipliers	4
3.3	Gaseous detectors	5
3.3.1	Operation modes of gaseous detectors	5
3.3.2	Drift chamber	5
3.4	Modern readout electronics	5
3.4.1	General aspects of detector readout	5
3.4.2	FPGA devices	5
3.4.3	Basics of logic circuit development	5
4	CMS Drift Tube system	6
4.1	Compact Muon Solenoid experiment	6
4.1.1	Large Hadron Collider	6
4.1.2	Experiment overview	6
4.2	Drift Tube system	6
4.2.1	Subdetector setup	6
4.2.2	Detector impact and performance	6
4.3	Phase-2 DT readout electronics	6
4.3.1	Motivation for upgrade	6
4.3.2	Minicrate electronics	6
4.3.3	HTG-940 backend	7
4.4	DT test stand in Aachen	7
4.4.1	MB1 DT chamber	7
4.4.2	Installation of Phase-2 minicrate	7

5	Triggerless Drift Tube data analysis and track reconstruction	8
5.1	Intra-channel timing calibration with testpulses	8
5.2	Drift cell hits	8
5.3	Superlayer hit clustering	8
5.4	Superlayer cluster fitting	8
5.5	Superlayer relative time alignment	9
5.6	Grouping of superlayer cluster fits	9
5.7	Global track building	9
5.8	Reconstructed muon tracks	9
5.9	Acceptance correction and angular distributions	9
6	SiPM-based scintillation detector	10
6.1	Detector setup	10
6.1.1	Mechanical structure	10
6.1.2	SiPM electronics	10
6.2	FPGA readout system	10
6.2.1	Input circuitry	10
6.2.2	FPGA firmware overview	10
6.2.3	Signal sampling with FPGA-internal comparators	10
6.2.3.1	Working principle	11
6.2.3.2	Voltage bias calibration	11
6.2.3.3	Input timing calibration	11
6.2.4	Online signal processing	11
6.2.4.1	Timestamp assignment	11
6.2.4.2	Data funneling	11
6.3	lpGBT Mezzanine PCB	11
6.3.1	lpGBT ASIC	12
6.3.2	Mezzanine board	12
6.3.3	Backend firmware modifications	12
7	Scintillator data analysis	13
7.1	SiPM noise scan	13
7.2	SiPM hits	13
7.3	SiPM coincidences	13
7.4	Scintillator strip coincidences	13
8	Combined measurement with both detectors	14
8.1	Combined detector setup	14
8.2	Temporal hit correlation	14
8.3	Spatial muon correlation	14
9	Summary and Outlook	15
	Appendix A Additional details	16
	Bibliography	17
	List of Acronyms	20

1 Introduction

- DT cosmic test stand in Aachen
- Scintillation detector self-developed at institute
- Read out both detectors together
- Create interface to DT chamber to later be used for trigger scintillators (for studies with DT chamber) or other detectors (using DT chamber triggerless reconstructed muons as reference)
- Requirement: Synchronous operation in ns range
- Use lpGBT chip to develop own readout system to interface Phase-2 DT readout
- lpGBT: Commonly used in HL-LHC upgrade project, ensures synchronous operation

2 The Standard Model of particle physics and cosmic muons

2.1 The Standard Model of particle physics

- Introduction to SM [1] [2]
- Particle zoo scheme [3]
- Particle physics books [4] [5] [6] [7] [8] [9] [10] [11]

2.1.1 Matter particles

- Fermi-Dirac statistics [12] [13]
- Pauli principle [14]
- 3 generations of matter
- Quarks
- Leptons

2.1.2 Force mediators

- Bose-Einstein statistics [15]
- Fundamental interactions and their mediator bosons
- Electroweak unification
- Chromodynamics and color confinement

2.1.3 Higgs boson

- Massive gauge bosons and fermions through spontaneous electroweak symmetry breaking [16] [17]
- Prediction of additional scalar boson
- Higgs boson discovery [18] [19]

2.1.4 Incompleteness and challenges

- Successful theory even in precision measurements
- Missing quantum gravitation
- Dark matter not explained [20]
- CP violation not allowed [21]
- Neutrino oscillations indicate massive neutrino(s), but massless in SM [22]

2.2 Cosmic muons

[23]

2.2.1 Cosmic radiation

- Origin of cosmic radiation [24]
- First observation of cosmic rays [25]
- Cosmic ray energy spectrum and particle types [26]

2.2.2 Atmospheric creation of muons

- Formation of muons from primary cosmic rays in Earth's atmosphere
- Discussion / criticism of terminology "cosmic muon"
- Energy spectrum and angular distribution of cosmic muons [27]

3 Principles of particle detection

3.1 Interaction of particles and matter

Particle detector detects the effects due to the particle interaction, not the particle itself.

3.1.1 Photon interactions

Very brief, since only interested in muon interactions.

- Photoeffect
- Compton effect
- Pair creation

3.1.2 Charged particle interactions

- Non-ionizing collisions
- Bethe-Bloch ionization
- Bremsstrahlung
- Other effects (transition radiation, Cherenkov radiation)

3.2 Scintillators and photon detection

- Scintillator: Photon emission due to energy deposition of particle
- Sensitive photon detection: PMT, SiPM (discussed in detail)

3.2.1 Scintillators

- Organic and anorganic scintillators
- Plastic scintillator (that is used in this thesis)

3.2.2 Silicon photomultipliers

- Geiger-Avalanche Photodiode array
- Sensitivity and performance
- Output signal shape
- Characteristic noise spectrum

3.3 Gaseous detectors

3.3.1 Operation modes of gaseous detectors

- Ionization chamber
- Proportional mode
- Nonlinear amplification mode
- Geiger mode

3.3.2 Drift chamber

- Proportional mode
- Constant electric field and drift velocity
- Correlation of charge travelling time and initial ionization position
- Drift chamber legacy in Aachen

3.4 Modern readout electronics

3.4.1 General aspects of detector readout

- Quantity to read out: E.g. voltage / current
- Discrimination of hits by amplitude
- Good time resolution
- More advanced: Resolution of signal amplitude and length, large number of channels

3.4.2 FPGA devices

- Brief history of FPGAs
- Fully programmable logic (difference to CPU)
- In particle physics: Fast, synchronized readout and processing of many channels

3.4.3 Basics of logic circuit development

- Description of logic circuitry implemented on FPGA
- Combinatorial logic
- Sequential / clocked logic
- Hardware Description Language
- IP cores

4 CMS Drift Tube system

4.1 Compact Muon Solenoid experiment

4.1.1 Large Hadron Collider

- Image of CERN accelerator complex
- Key facts: Luminisity, collision center-of-mass energy

4.1.2 Experiment overview

- CMS rendering
- Brief listing of all subdetectors

4.2 Drift Tube system

4.2.1 Subdetector setup

- DT chambers in CMS: Wheels, sectors, stations
- Components of one DT chamber: 2-3 superlayers, 4 layers, drift cells
- Gas mixture, HV values

4.2.2 Detector impact and performance

- Relevance of muon detection (e.g. dimuon Higgs decay)
- DT performance (efficiency and active channels)
- DT system resolution

4.3 Phase-2 DT readout electronics

4.3.1 Motivation for upgrade

- High-Luminosity LHC
- Increased pileup and tightened radiation environment

4.3.2 Minicrate electronics

- Minicrate location on chamber
- Minicrate functionalities
- On-Board Electronics for Drift Tubes boards

4.3.3 HTG-940 backend

- Prototype backend (AMD Xilinx Virtex UltraScale+ FPGA)
- Production backend: Serenity and BMTL1 (not available yet)
- Backend firmware (mention own modifications)

4.4 DT test stand in Aachen

4.4.1 MB1 DT chamber

- MB1 DT chamber from production in 2000s
- Reactivated infrastructure

4.4.2 Installation of Phase-2 minicrate

- Minicrate on DT chamber
- Detailed setup and channel mapping

5 Triggerless Drift Tube data analysis and track reconstruction

5.1 Intra-channel timing calibration with testpulses

- Importance of timing calibration in DT
- Refer to CMS DT calibration procedure
- Show testpulse response
- Discuss different testpulse cable length in theta SL
- Show derived timing corrections for individual channels

5.2 Drift cell hits

- Show hit occupancies of all wires
- Give list of masked (noisy) and dead wires
- Discuss rate
- Highlight and discuss feature of secondary hits in range 0-400 ns
- Apply dead time for individual channels

5.3 Superlayer hit clustering

- Triggerless reconstruction: Need to consider all recorded hits as potential muon signature
- Cluster hits within maximum drift time window
- Pattern rate discussion
- Compare to rates of fake patterns (to estimate noise influence)

5.4 Superlayer cluster fitting

- Assume constant drift velocity, straight muon track
- Perform linear fit for all laterality combinations with parameters: Arrival time, position in cell, track inclination
- Select laterality combination with best goodness of fit
- Discuss fit results
- Apply appropriate goodness of fit cut

5.5 Superlayer relative time alignment

- Mention relative alignment problem of OBDTs
- Show double peak in time difference between SL fits
- Establish relative time correction

5.6 Grouping of superlayer cluster fits

- Clustering algorithm creates multiple clusters in case of additional nearby hits
- Cope this by grouping together clusters within certain time window within same superlayer

5.7 Global track building

- Select cluster groups within time window from all superlayers
- Only use cluster groups with no ambiguity (only one cluster per SL)
- Ensure matching tracks in both phi superlayers
- Calculate 3D track from 2D projections from theta and phi superlayers

5.8 Reconstructed muon tracks

- Discuss muon rate
- Show angular distributions (with solid angle correction)

5.9 Acceptance correction and angular distributions

- Establish geometric simulation of DT chamber with simulated muon tracks
- Derive (rudimentary) geometric acceptance correction from simulation
- Apply acceptance correction to data
- Compare to expected cosmic muon angular distributions

6 SiPM-based scintillation detector

6.1 Detector setup

Mention self-development of detector at institute [28] [29].

6.1.1 Mechanical structure

- 2x 16 plastic scintillator strips with wavelength-shifting fibers
- Dual SiPM readout for each strip

6.1.2 SiPM electronics

- SiPM front-end board with preamplifier
- Backbone of detector with bias voltage supply
- Raspberry Pi for backbone control

6.2 FPGA readout system

- Mention completely new hardware and firmware development in scope of this thesis, based on previous experience and efforts of our group
- AMD Xilinx UltraScale+ ZynqMPSoC
- Control hardware and firmware from Petalinux running on ARM core of ZynqMPSoC
- Show scheme / overview of readout system

6.2.1 Input circuitry

- Synchronous testpulse source
- SiPM signal input and testpulse input (AC coupled)
- Individually selectable threshold for all channels
- (Show screenshot from SPICE simulation)

6.2.2 FPGA firmware overview

Mention fully self-developed readout firmware.

6.2.3 Signal sampling with FPGA-internal comparators

- Comparators for input signal discrimination, however no conventional, dedicated hardware but FPGA-internal comparators used.
- Use FPGA-internal blocks for deserialization to increase time resolution

6.2.3.1 Working principle

- FPGA-internal comparators (intended for differential signalling)
- Novel concept: Use FPGA-internal comparators with analog signals [30] [31] [32]
- Voltage limits of inputs
- Deserializers for higher time resolution

6.2.3.2 Voltage bias calibration

- FPGA-internal comparators (on FPGA silicon die) are subject to production uncertainties
- Unfolds in voltage bias on comparator switching point
- Scan input voltage range while keeping other comparator input constant to determine switching point of comparator (Metastability calibration)
- Account for measured biases when setting thresholds

6.2.3.3 Input timing calibration

- Bank-specific calibration of deserializers on each power-cycle leads to non-deterministic signal delays
- Use simultaneous testpulses to measure these delays
- Use delay module in firmware to correct for these effects already on the device
- Mention ringing of testpulse signal in circuit, but explain that no negative effect on calibration

6.2.4 Online signal processing

6.2.4.1 Timestamp assignment

- Hit arrival time translated to (absolute) timestamp
- Same format as ODBT hits

6.2.4.2 Data funneling

- Limited bandwidth: Can not transmit hits of all channels together
- However do not want to lose hits
- Solution: Introduce individual buffer (FiFo) for all channels
- Dynamically cycle through and activate outputs of different channel buffers
- Reduces possible hit rate compared to no funnel

6.3 lpGBT Mezzanine PCB

- Hardware interface between scintillator readout system and DT backend
- ODBTs use lpGBT as well therefore compatible format
- Synchronous clock (with backend) supplied by lpGBT chip

6.3.1 lpGBT ASIC

- Low-power GigaBit Transceiver ASIC developed centrally by CERN
- Key facts: Data transfer rate, clock recovery
- Optical fiber interface to backend
- Electrical link (eLink) interface to frontend FPGA
- Control interface via optical fiber or wired I2C

6.3.2 Mezzanine board

- Connects eLinks between lpGBT and FPGA
- (Show bit error rates of PRBS7 test)
- Show recovered 40 MHz clock
- Hosts optical transceiver to convert signals to/from optical fiber

6.3.3 Backend firmware modifications

- Modified data flow in DT backend to accomodate reception of data via slow control link from the lpGBT Mezzanine (OBDT has dedicated trigger links for data)
- (Show received counter data by backend)

7 Scintillator data analysis

7.1 SiPM noise scan

- Scan of SiPM rates for different thresholds with developed readout system
- Show resulting curve
- Discuss visible exponential (expected) behavior
- Aliasing effects (too coarse scanning steps) hide expected steps in spectrum

7.2 SiPM hits

- Show single SiPM occupancies
- Discuss observed secondary pulses: Ringing in circuit or SiPM after-pulses

7.3 SiPM coincidences

- Coincidence of two SiPMs of same scintillator strip
- Discuss strip rate
- Show time difference between both SiPMs, discuss width / time localization
- Compare to expected fake coincidence rate

7.4 Scintillator strip coincidences

- Coincidence of two strips of two detector layers (create "pixels")
- Discuss pixel rate
- Show time difference between both strips
- Compare to expected fake coincidence rate

8 Combined measurement with both detectors

8.1 Combined detector setup

- Overview schematic of combined detector and readout setup
- Highlight shared backend and synchronized clocks

8.2 Temporal hit correlation

- Group scintillator pixel hits with reconstructed DT tracks in time window
- Discuss rate of correlated events (compared to scintillator pixel hits before correlation)
- Show time difference between DT and scintillator
- Discuss distribution, highlight the synchronization

8.3 Spatial muon correlation

- Show cumulated reconstructed DT tracks for "matched" hits between scintillator and DT
- Highlight good geometric agreement
- Mention need for alignment analysis (which in principle would be possible with this data)

9 Summary and Outlook

Summary:

- Completely new SiPM readout system developed
- Incorporating modern FPGA data acquisition and FPGA-internal comparators for analog signal discrimination
- Adapted and developed new calibration procedures necessary for FPGA readout system
- Developed lpGBT Mezzanine to interface DT backend
- Developed FPGA firmware for readout and interfacing the lpGBT
- Successful modification of DT backend firmware to interface developed readout system
- Installed and commissioned Phase-2 minicrate with DT chamber at test stand in Aachen
- Framework to analyze DT and scintillator data together
- Implemented own triggerless reconstruction code which follows the CMS DT reconstruction
- Successful correlation of measured DT tracks and scintillator hits in space and time

Outlook:

- Understand open problems
- Implement coincidence IP core (currently still in development)
- Use system to perform various studies with DT chamber, scintillator or even other detectors connected to the developed readout system

Appendix

A Additional details

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List of Acronyms

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